

WebXR, CAVEs, and the Balance of XR Platform Agnosticity Versus Performance in Immersive Scientific Visualization

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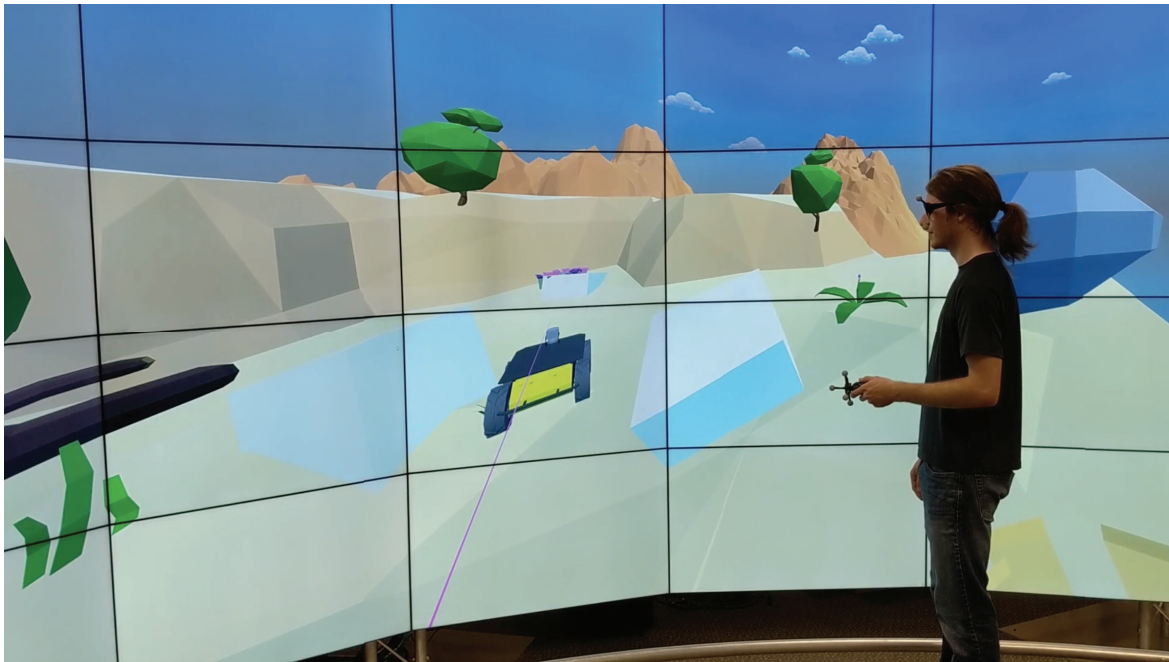


Figure 1: A simulation for human-robot interaction on the UMBC Mechdyne ARC system. This simulation, built in Unity specifically for this display environment, is an example of a low-polygon project which could have been built with WebXR and run on more devices.

ABSTRACT

This position paper explores the potential of WebXR compatibility for increased impact from large-scale visualization environments housed in immersive visualization laboratories. In a rapidly evolving landscape of highly performant extended reality (XR) head-mounted displays (HMDs), the ability to develop software which can meaningfully utilize relative strengths of different visualization tools is paramount. We explore how use of standards such as WebXR for broadly platform-agnostic immersive visualization compares to traditional methods of more tailored development, noting costs and benefits of each approach.

Index Terms: Human-centered computing—Interaction paradigms—Virtual reality; Human-centered computing—Interaction paradigms—Web-based interaction; Human-centered computing—Visualization systems and tools

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1 INTRODUCTION

Consumer-grade extended reality head-mounted displays (HMDs) are providing rapidly improving functionality at the relatively low cost of hundreds to thousands of U.S. dollars per unit. On the one hand, this cost efficiency invites us to revisit the value proposition of immersive visualization systems (e.g., modern CAVEs), which often require an initial investment in the hundreds of thousands of dollars and ongoing investments in specialized staffing and upkeep, as is typical of university-scale core facilities [2]. On the other hand, the advent of affordable, high-quality HMDs is leading to the creation of a wider range of software and content which could potentially be ported to CAVE-like systems, increasing their impact and value proposition. In this position paper, we argue for community developments which improve the compatibility of CAVE-like systems with platform-agnostic XR software, especially software built with technologies such as WebXR for primary use with HMDs.

For immersive visualization technologies of all styles and price points, there are a wide range of applications such as gaming; fitness; social interaction; education and training; productivity; and scientific research. Within these broad categories, more specific use cases may have requirements which are only achievable with certain hardware systems. For example, there is tremendous variation in the extent to which different use cases require systems to provide various levels of resolution; frame rate; mobility; co-presence; blending into physical reality; processing power; and Internet connectivity. Notably, there

are many scenarios (including in scientific visualization) where ease of adoption by a large number of XR novices is a priority, and where there are no significant requirements for resolution; blending into physical reality; processing power; or co-presence. Scenarios of interest include those where immersion is used to enable scientific reasoning about a scene with relatively simple 3D structure and potentially complex 2D textures. Such scenes can be presented with relatively minimal hardware such as phones, tablets, computer screens, and the cheapest HMDs. When such scenes are displayed in CAVEs, they can be used to train users for individual workflow tasks which they will subsequently perform in an HMD; for education (e.g., using visualizations to educate K-12 and university classes); and to provide a higher caliber version of an HMD experience, perhaps allowing for more details to be noticed and included in a scientific user's mental model. Our position is inspired by our own work in developing a WebXR visualization of the Greenland Ice Sheet for scientific use by glaciologists [17, 18].

2 RELATED WORK

2.1 CAVEs and Room-Scale Display Environments

For several decades, large institutions such as universities and government agencies have invested in room-sized immersive environments, providing stereoscopic, parallax-enabled visualizations to one or more users. Cave Automatic Virtual Environments (CAVEs) [4] were first developed by the University of Illinois, Chicago (UIC) in 1992, using projectors on some of the walls, floor, and/or ceiling of a cube-shaped room. This design was replicated at many universities and government agencies. In 2012, the same UIC group which created the original CAVE modernized the design as CAVE2 [6]. A CAVE2 system consists of 72 stereoscopic 1080p LCD screens. Columns of four screens each form a near-complete circle that is over 20 feet in diameter. The CAVE2 design, licensed to Mechdyne, has also been broadly adopted. Due to budgetary and space constraints, some institutions have built partial CAVE2 systems called Mechdyne ARCs, such as our own six-column, 24-screen display [11]. In addition to displays and 3D tracking, these systems include rack-mounted computing resources with multiple nodes, high-end GPUs, and memory, enabling high performance for complex scenes involving 50 to 150 million pixels at high frame rates.

Systems of this scale and resolution allow for work which is not achievable in a consumer-grade standalone headset. Accordingly, many of the projects which have been built for CAVEs, CAVE2s, ARCs, and other large-scale systems are designed to work only with that specific type of system. Because of variations in the designs even within each of these categories (e.g., CAVEs with different numbers of walls, and ARCs with different numbers of columns), most projects are tailored to work only at a single site. However, there have been some projects which are designed primarily with such systems in mind, yet are built to be versatile enough to work at many sites and even across other types of display systems. Virtual Reality Peripheral Network (VRPN) [19] provides an abstraction layer between VR-oriented hardware and VR applications, simplifying and accelerating application development at a wide range of visualization facilities, including our own. Another approach to platform independence within the range of visualization facilities is Smart Amplified Group Environment (SAGE) [8], which allows for "hyper screen sharing" on wall-style displays, with each user independently controlling their own cursors and keyboards in a shared environment of both native SAGE applications and projected laptop screens.

As HMD headsets have become increasingly performant, some in the research community have argued that CAVEs have limited value proposition relative to their significantly higher costs. In considering the future of their visualization facility at Villanova University, Klassner et al [9] note that some recent studies which make this argument are focused on a relatively narrow set of uses, such as

Elor et al's HMD-vs-CAVE analysis [5]. Elor et al gave participants a task which took only a few minutes to complete, and Klassner notes that such tasks do not take into account considerations like eye strain from longer-term (e.g., 20 minute) use of HMDs [22] or face-to-face human interactions. Klassner asks: "Will centers that use large-scale facilities like CAVEs need to replace them with racks full of headsets?" and answers: "I believe based on the experience of my Center that the answer ultimately rests on what kind of research and educational work a center plans on performing and what use cases a center wants to support" [9].

2.2 WebXR, OpenXR, and Scientific Visualization

Software development for HMDs and other mobile devices faces a parallel challenge, perhaps to a lesser extent. HMD manufacturers have their own SDKs, but more importantly, each headset and controller has its own distinct set of input and display capabilities. To allow for a single pathway for XR development across many platforms, the World Wide Web Consortium (W3C) has developed and maintains the WebXR Device API standard. This W3C's Immersive Web Working Group includes representation from major stakeholders in the XR industry, including Meta, Google, Apple, Samsung, Microsoft, and Magic Leap [21]. While Unity, Unreal, and other game development engines can be used to compile tailored executables for different XR platforms, WebXR provides the additional benefit of allowing users to launch an XR experience through most common web browsers, avoiding the need for content creators or users to navigate app stores or to install software.

Software has been developed to run natively on specific HMDs for scientific and data visualization in a wide range of disciplines (by others and ourselves), in domains such as biology [15], cybersecurity [10], meteorology [7], robotics [13], and glaciology [1]. Relatively few published works (e.g., ProteinVR [3] and our own work in glaciology [17, 18]) use WebXR or related web-based, platform-agnostic technologies for immersive scientific and technical visualizations. In addition to the few scientific visualization projects which use these tools, new software libraries and related studies aim to support others in the development of WebXR-based data visualizations [12, 14, 16]. While these WebXR visualizations and supporting tools are built with XR platform agnosticism in mind, they are still built with HMDs and 2D screens as the presumed viewing modalities.

A related standard to WebXR is OpenXR by the Khronos group [20], a consortium with many of the same industry members as the W3C's Immersive Web Working Group. While the WebXR API enables web-based XR experiences, the OpenXR standard is focused on enabling XR applications to run natively on a variety of XR devices. Because our focus includes ease of portability and adoption by a scientific user community, our immediate interest is in XR experiences which can be launched from a browser without requiring software installations on a per-application basis. Nonetheless, it is worth noting that OpenXR offers similar potential for portability of existing VR experiences to CAVE-like systems, so the development of an OpenXR-based bridge to CAVE systems is also worthy of further exploration.

3 OUR POSITION AND WORK IN PROGRESS

We propose that certain 3D scientific visualizations are well-positioned for simultaneous deployment in everything from 2D screens (phones and computers), XR HMDs, and CAVE-like large-scale facilities.

3.1 WebXR Support in CAVE-like Systems

WebXR-enabled web applications typically start by showing a flat view of a 3D scene within a web browser. Such a web browser could be on a phone, tablet, computer, or floating as a flat simulated screen in a virtual environment (Figure 2). Users launch immersive sessions

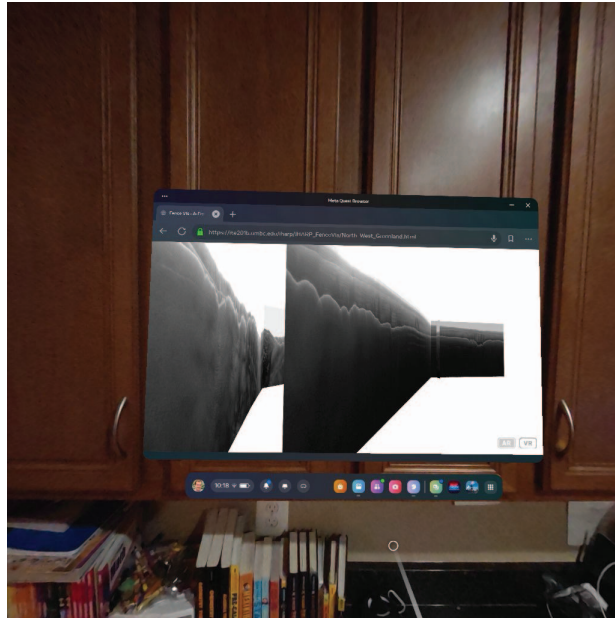


Figure 2: Fence diagram of ice-penetrating radar images of the Greenland Ice Sheet in the Quest 3's mixed reality web browser before launching an immersive session.

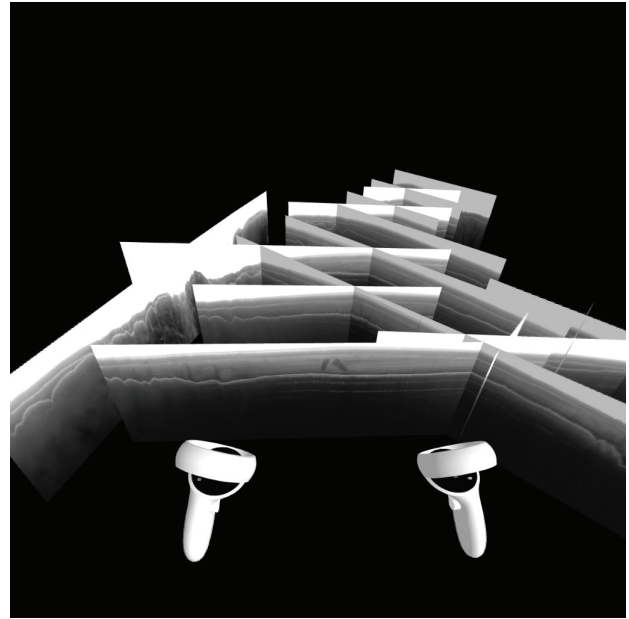


Figure 3: The same fence diagram from Figure 2 after launching an immersive session.

(Figure 3) by clicking a button within this view. While most CAVE systems are running a Windows or Linux operating system and have web browsers installed, these browsers hand the immersive sessions to downstream software to manage the XR display and interactions. For example, if a computer running a CAVE also has a tethered HMD device attached, the interactive session could be launched without difficulty to be displayed in that HMD. However, the display and tracking systems of CAVEs do not present themselves to the CAVE computer's operating system as a generic virtual reality device, and thus are not recognized as a valid display destination for WebXR content.

We are currently in the initial stages of planning and developing software which will enable our CAVE-like system (a Mechdyne ARC-6) to recognize its XR components as though they were an attached XR device, akin to an HMD. Our system's off-the-shelf Advanced Realtime Tracking (ART) system will provide the coordinates of the position and orientation of a primary user's 3D glasses through VRPN [19], while the quad-buffered stereo, which feeds our 24 monitors, will be used to render the separate views for the left and right eyes. Due to the geometry of our system, this will require a total of 12 virtual cameras, one pair for each of the six columns of stereoscopic monitors. The positions and orientations of controllers will also be tracked by the ART system, and the buttons on these controllers will be transmitted via VRPN and mapped to WebXR inputs. Because WebXR can deal with controllers abstractly, no system-specific changes will be required in the WebXR-based visualization program, and most WebXR-based sites will work, unmodified, using this same bridge to our CAVE-like system. Our intent is to develop this bridge such that a configuration file would allow other CAVE-like systems to work with WebXR content by specifying their own display geometries and 3D tracking systems.

3.2 Identification of Appropriate Use Cases

Not all scientific visualizations are well-positioned for deployment on all 2D and XR display technologies. From our own work in developing WebXR visualizations of fence diagrams of the Green-

land Ice Sheet, we believe there is a set of 3D scenes particularly well-suited for meaningful display on the full gamut of visualization hardware. Another prior project in our lab, shown in Figure 1, involved a simple landscape for simulated human-robot interaction. Both projects involved low 3D complexity, allowing for deployment on rudimentary HMDs and phones with limited computational power. They may have complex 2D textures on these 3D surfaces, allowing for a visually complex environment despite having a low polygon count. These scenes should have a clear value proposition for immersion-based visual analysis, with 3D information and freedom of movement enabling the user to draw insights which would not be possible from a single image.

When these conditions are met by a use case, the versatility afforded by deployment onto such a wide range of devices offers significant potential. The portability and ubiquity of phones, tablets, and laptops allows for broad adoption of the visualizations by scientific domain experts who may lack access to (or enthusiasm for) more immersive technologies. For those with access to HMDs, the ability to step into the 3D environment and move one's perspective and controllers with six degrees of freedom allows for a deeper and more natural engagement with the same data. Showing the same scenes on a CAVE2, where many users can easily put on 3D glasses and follow the movements and perspective of a primary user, enables a level of multi-person engagement and discussion unachievable by the other display technologies, and is particularly well-suited for collaborative analysis, training, education, and public science outreach. The larger displays also afford orders of magnitude higher resolution, which may enable discovery through accurate presentation of more detailed imagery. Critically, if CAVE systems are supported by a software bridge which allows them to show WebXR content without tailored development effort for each visualization project, this can then all be accomplished with no additional effort by software engineers, so long as they have made the design choice to use WebXR.

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